Session-6-exercises.R

```
###########################
### Session 6 Exercises ###
###########################


# Load the necessary packages
library(tidyverse)
library(readxl)

# Warmup with HR data

dataHR <- read_excel("dataHR_labeled.xlsx")

# The first 6 rows are given by head():
head(dataHR)

## # A tibble: 6 × 6
##   sex    workexp   age jobtitle        wage edu
##   <chr>    <dbl> <dbl> <chr>          <dbl> <chr>
## 1 male         9    39 administrative  2370 secondary
## 2 male        16    31 administrative  3125 undergraduate
## 3 female       8    53 professional    2620 secondary
## 4 male         6    35 administrative  2270 secondary
## 5 female       7    38 professional    3315 postgraduate
## 6 female       8    59 manager         3375 postgraduate

# The last 6 rows are given by tail():
tail(dataHR)

## # A tibble: 6 × 6
##   sex   workexp   age jobtitle        wage edu
##   <chr>   <dbl> <dbl> <chr>          <dbl> <chr>
## 1 male        8    40 administrative  1910 secondary
## 2 male       10    36 administrative  2540 secondary
## 3 male        1    27 administrative  2345 secondary
## 4 male        2    43 administrative  2510 secondary
## 5 male        3    34 administrative  2255 secondary
## 6 male        4    30 administrative  2535 secondary

# A summary of descriptive statistics that works fine for numeric variables
and factors.
summary(dataHR)

##      sex               workexp           age          jobtitle
##  Length:500         Min.   : 1.00   Min.   :27.0   Length:500
##  Class :character   1st Qu.: 6.00   1st Qu.:37.0   Class :character
##  Mode  :character   Median :10.00   Median :46.0   Mode  :character
##                     Mean   :10.02   Mean   :45.5
##                     3rd Qu.:13.00   3rd Qu.:52.0
##                     Max.   :30.00   Max.   :70.0
```

```
##       wage          edu
##  Min.   :1890   Length:500
##  1st Qu.:2555   Class :character
##  Median :2950   Mode  :character
##  Mean   :3175
##  3rd Qu.:3550
##  Max.   :7465
```

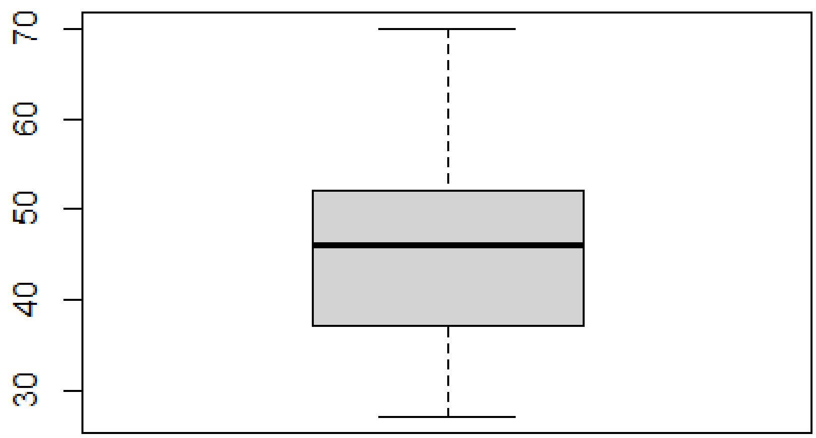
```r
dataHR$sex <- as.factor(dataHR$sex)

summary(dataHR)
```

```
##      sex         workexp           age          jobtitle              wage
##  female:212   Min.   : 1.00   Min.   :27.0   Length:500         Min.   :1890
##  male  :288   1st Qu.: 6.00   1st Qu.:37.0   Class :character   1st Qu.:2555
##               Median :10.00   Median :46.0   Mode  :character   Median :2950
##               Mean   :10.02   Mean   :45.5                      Mean   :3175
##               3rd Qu.:13.00   3rd Qu.:52.0                      3rd Qu.:3550
##               Max.   :30.00   Max.   :70.0                      Max.   :7465
##      edu
##  Length:500
##  Class :character
##  Mode  :character
##
##
##
```

```r
#Look at the difference

dataHR$jobtitle <- ordered(dataHR$jobtitle, levels=c('administrative',
'professional', 'manager'))
dataHR$edu <- ordered(dataHR$edu, levels=c('secondary', 'undergraduate',
'postgraduate'))

# Filtering and subsetting is possible:
summary(dataHR[dataHR$wage>5000,])
```

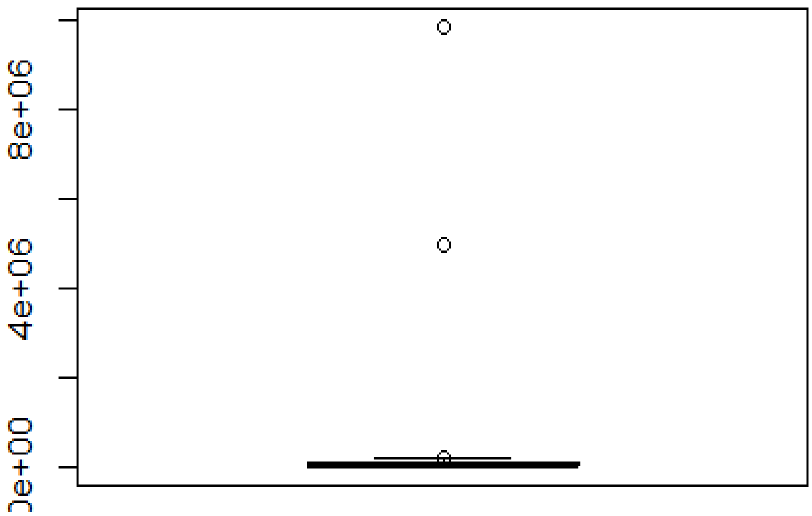
```
##      sex        workexp          age                   jobtitle         wage
##  female:11   Min.   :14.00   Min.   :36.00   administrative: 0   Min.   :5005
##  male  : 9   1st Qu.:17.75   1st Qu.:48.75   professional  : 3   1st Qu.:5189
##              Median :23.00   Median :53.50   manager       :17   Median :5658
##              Mean   :22.70   Mean   :54.05                       Mean   :5727
##              3rd Qu.:27.25   3rd Qu.:60.75                       3rd Qu.:5944
##              Max.   :30.00   Max.   :68.00                       Max.   :7465
##             edu
##  secondary    :12
##  undergraduate: 8
##  postgraduate : 0
##
##
##
```

```r
# Frequency distribution for one variable that works really fine for
character variables:
table(dataHR$sex)

## female   male
##    212    288

# A boxplot (box-and-whiskers plot) can be useful.
boxplot(dataHR$age)
```

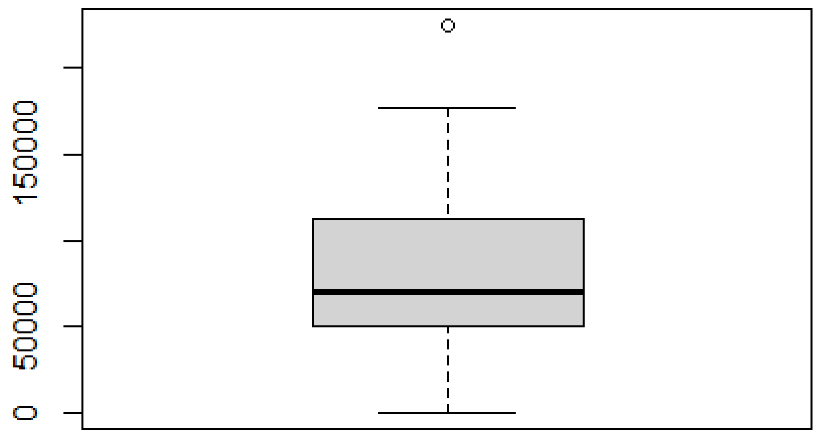



```r
# Frequency distribution (cross-tabulation) for two variables:
table(dataHR$sex, dataHR$edu)

##          secondary undergraduate postgraduate
##   female        93            85           34
##   male         192            71           25

# Proportion (percentages) for a table (% of grand total, % of row total, %
of column total):
prop.table(table(dataHR$sex, dataHR$edu))

##          secondary undergraduate postgraduate
##   female     0.186         0.170        0.068
##   male       0.384         0.142        0.050

prop.table(table(dataHR$sex, dataHR$edu), 1)

##           secondary undergraduate postgraduate
##   female 0.43867925    0.40094340   0.16037736
##   male   0.66666667    0.24652778   0.08680556

prop.table(table(dataHR$sex, dataHR$edu), 2)

##          secondary undergraduate postgraduate
##   female 0.3263158     0.5448718    0.5762712
##   male   0.6736842     0.4551282    0.4237288
```

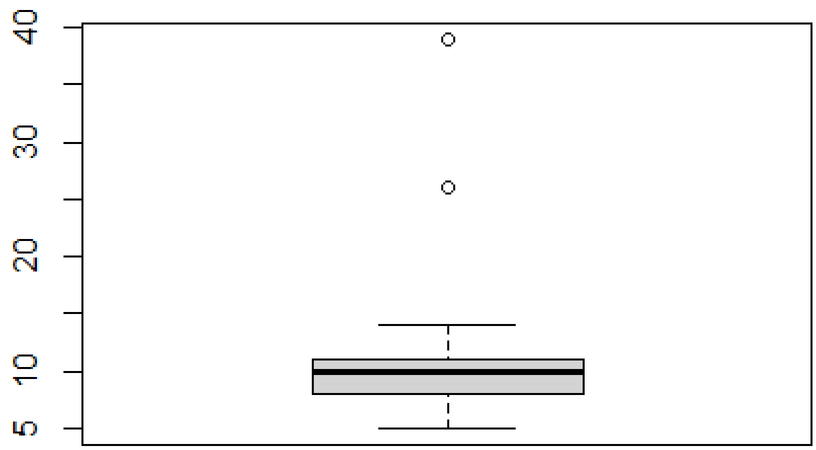
```
# Who earns more? Really?

# Load the data file
whitehouse <- read_csv("whitehouse.csv", col_types="ccncci")

# Look at a boxplot of salary data
boxplot(whitehouse$Salary)
```



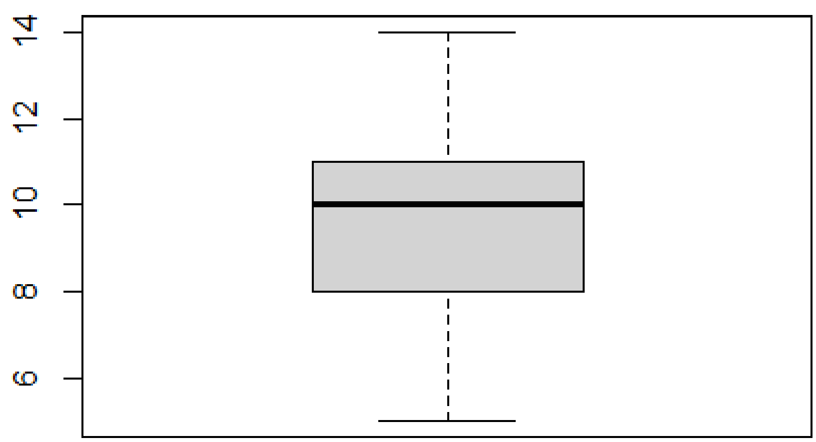

```
# Find salaries over $1,000,000
whitehouse %>%
  filter(Salary>1000000)

## # A tibble: 2 × 6
##   Name                 Status    Salary `Pay Basis` `Position Title`    Year
##   <chr>                <chr>      <dbl> <chr>       <chr>              <int>
## 1 Case, Michael A.     Employee 5000000 Per Annum   SENIOR WRITER       2011
## 2 Blair, Patricia A.   Employee 9866900 Per Annum   CHIEF CALLIGRAPHER  2015

# Set salaries over $1,000,000 to NA
whitehouse <- whitehouse %>%
  mutate(Salary=ifelse(Salary>1000000, NA, Salary))

# Rerun the box plot
boxplot(whitehouse$Salary)
```



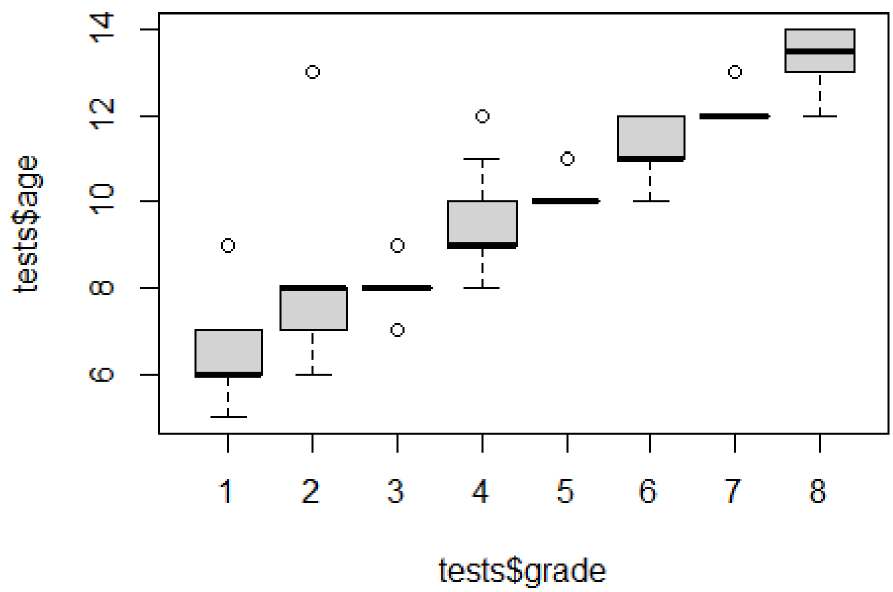

```
# Find the salary over $200,000
whitehouse %>%
  filter(Salary>200000)

## # A tibble: 1 × 6
##   Name              Status   Salary `Pay Basis` `Position Title`       Year
##   <chr>             <chr>     <dbl> <chr>       <chr>                 <int>
## 1 Wheeler, Seth F. Detailee 225000 Per Annum   SENIOR POLICY ADVISOR  2013

# Being retired in the primary school?
# Load the data file
tests <- read_csv("testscores.csv")

## Rows: 217 Columns: 4
## ── Column specification ────────────────────────────────────────────────────────
## Delimiter: ","
## dbl (4): studentID, age, grade, testScore
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.

# Look at summary statistics
summary(tests)

##    studentID          age             grade       testScore
##  Min.   :10001   Min.   : 5.000   Min.   :1.0   Min.   : 59.00
##  1st Qu.:10055   1st Qu.: 8.000   1st Qu.:3.0   1st Qu.: 75.00
##  Median :10109   Median :10.000   Median :4.0   Median : 83.00
##  Mean   :10109   Mean   : 9.871   Mean   :4.3   Mean   : 82.43
##  3rd Qu.:10163   3rd Qu.:11.000   3rd Qu.:6.0   3rd Qu.: 89.00
##  Max.   :10217   Max.   :39.000   Max.   :8.0   Max.   :114.00

boxplot(tests$age)
```




```r
# Investigate outliers
tests %>%
  filter(age>15)

## # A tibble: 2 × 4
##   studentID   age grade testScore
##       <dbl> <dbl> <dbl>     <dbl>
## 1     10115    39     2        91
## 2     10116    26     7        70

# Remove those two cases and reexamine boxplot
tests <- tests %>%
  mutate(age=ifelse(studentID==10115, 7, age)) %>%
  mutate(age=ifelse(studentID==10116, 12, age))

boxplot(tests$age)
```



```r
# But look at it by grade level
boxplot(tests$age~tests$grade)
```




```r
# Homeless? Mama hotel?
# Load the data file
residents <- read_csv("residents.csv", col_types='iillll')

# Look at a summary of the data
summary(residents)
```

```
##     personID           age          employed        ownsHome
##  Min.   :   1.0   Min.   :18.00   Mode :logical   Mode :logical
##  1st Qu.: 516.5   1st Qu.:31.00   FALSE:424       FALSE:837
##  Median :1032.0   Median :42.00   TRUE :1639      TRUE :1226
##  Mean   :1032.0   Mean   :42.22
##  3rd Qu.:1547.5   3rd Qu.:54.00
##  Max.   :2063.0   Max.   :65.00
##  rentsHome        ownsCar
##  Mode :logical   Mode :logical
##  FALSE:1224      FALSE:635
##  TRUE :839       TRUE :1428
##
```

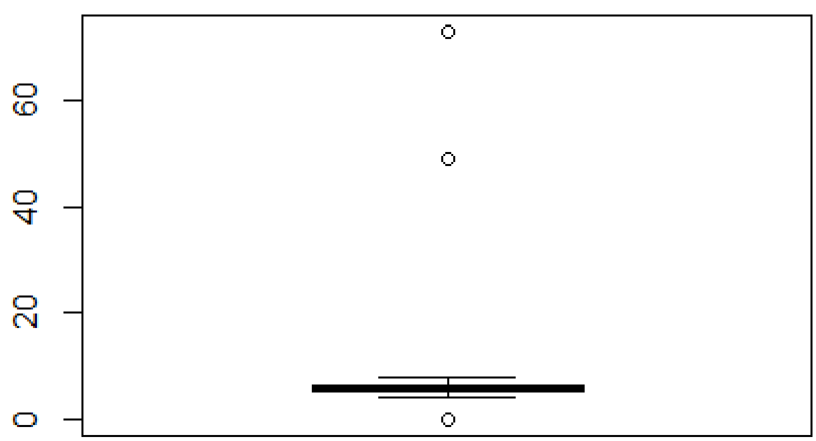
```r
# Find unusual cases
residents %>%
  filter(ownsHome==rentsHome)
```

```
## # A tibble: 6 × 6
##   personID   age employed ownsHome rentsHome ownsCar
##      <int> <int> <lgl>    <lgl>    <lgl>     <lgl>
## 1       17    55 TRUE     FALSE    FALSE     TRUE
## 2      134    19 TRUE     TRUE     TRUE      FALSE
## 3      203    43 FALSE    TRUE     TRUE      TRUE
## 4      382    60 FALSE    TRUE     TRUE      TRUE
## 5     1269    56 TRUE     TRUE     TRUE      TRUE
## 6     1902    31 TRUE     FALSE    FALSE     FALSE
```

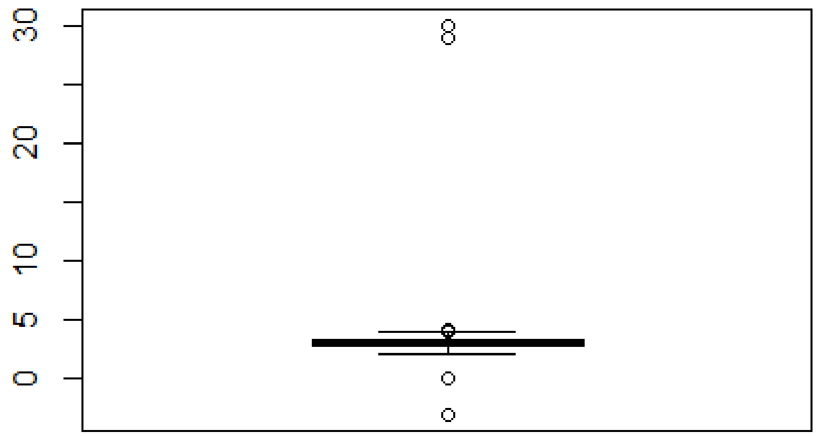
```r
################################################################

#Exercise 6.1
# a) Load the “dirty petal.csv” file.
iris <- read_csv("dirty petal.csv")
```

```
## Rows: 150 Columns: 5
## ── Column specification
────────────────────────────────────────────────────────
## Delimiter: ","
## chr (1): Species
## dbl (4): Sepal.Length, Sepal.Width, Petal.Length, Petal.Width
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

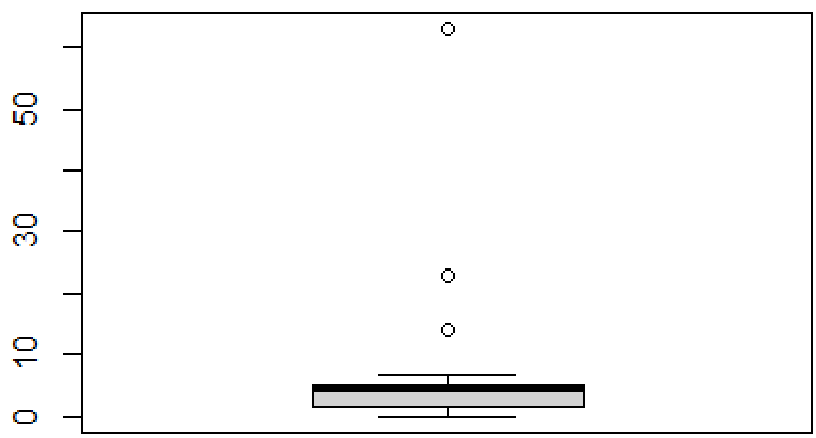
```r
# b) Search for outliers and propose a solution for them.
boxplot(iris$Sepal.Length)
```



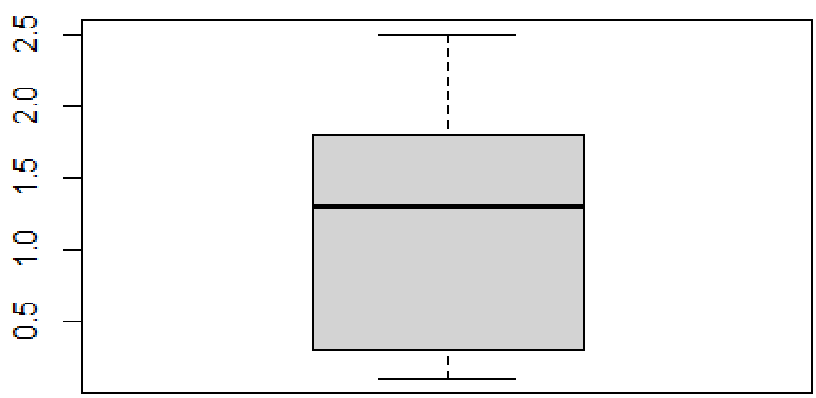
```r
boxplot(iris$Sepal.Width)
```



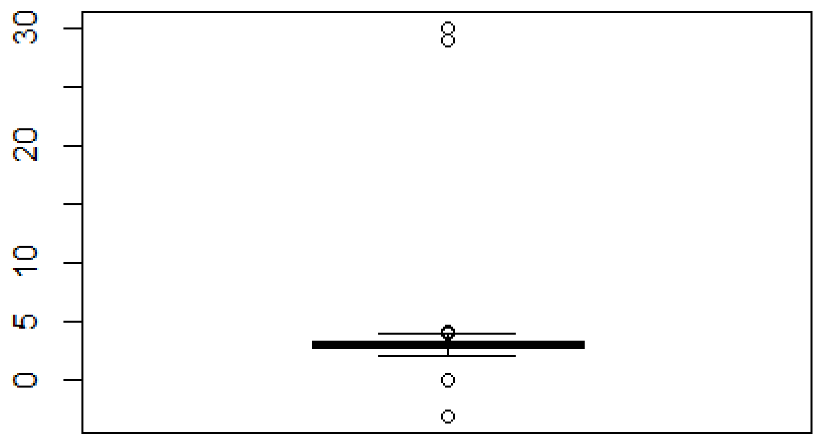

```r
boxplot(iris$Petal.Length)
```




```r
boxplot(iris$Petal.Width)

## Warning in bplt(at[i], wid = width[i], stats = z$stats[, i], out =
## z$out[z$group == : Outlier (Inf) in boxplot 1 is not drawn
```

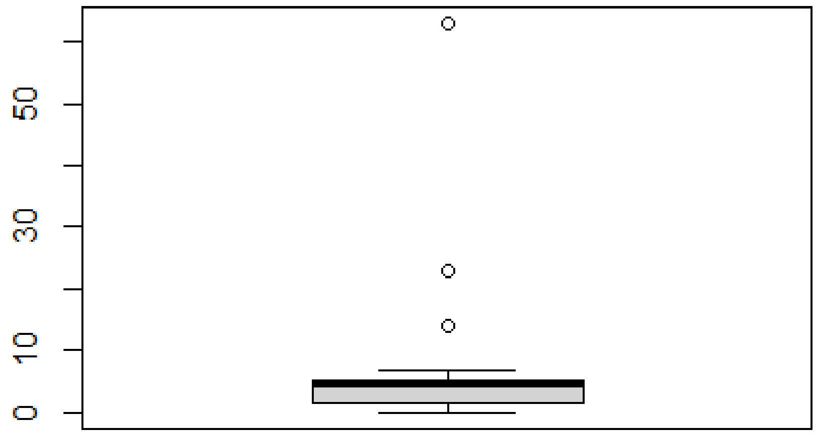


```r
summary(iris)

##   Sepal.Length     Sepal.Width      Petal.Length    Petal.Width
##  Min.   : 0.000   Min.   :-3.000   Min.   : 0.00   Min.   :0.1
##  1st Qu.: 5.100   1st Qu.: 2.800   1st Qu.: 1.60   1st Qu.:0.3
##  Median : 5.750   Median : 3.000   Median : 4.50   Median :1.3
##  Mean   : 6.559   Mean   : 3.391   Mean   : 4.45   Mean   :Inf
##  3rd Qu.: 6.400   3rd Qu.: 3.300   3rd Qu.: 5.10   3rd Qu.:1.8
##  Max.   :73.000   Max.   :30.000   Max.   :63.00   Max.   :Inf
##  NA's   :10       NA's   :17       NA's   :19      NA's   :12
##    Species
##  Length:150
##  Class :character
##  Mode  :character
##


# 2 or 3 outliers in Sepal.Length
filter(iris, Sepal.Length >10)

## # A tibble: 2 × 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##          <dbl>       <dbl>        <dbl>       <dbl> <chr>
## 1           73          29           63          NA virginica
## 2           49          30           14           2 setosa

#For both of them Sepal.Length is multiplied by 10
# But seemingly the same problem for Sepal.Length and Petal Length
iris <- iris %>%
  mutate(Sepal.Length=ifelse(Sepal.Length>10, Sepal.Length/10, Sepal.Length))

filter(iris, Sepal.Length < 1)

## # A tibble: 1 × 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##          <dbl>       <dbl>        <dbl>       <dbl> <chr>
## 1            0          NA          1.3         0.4 setosa

#Maybe the Sepal was completely missing, only the petal measured.
#Replace by the mean of the species, or drop.
iris <- iris[iris$Sepal.Length>0|is.na(iris$Sepal.Length),]

# 4 outliers in Sepal.Width
boxplot(iris$Sepal.Width)
```



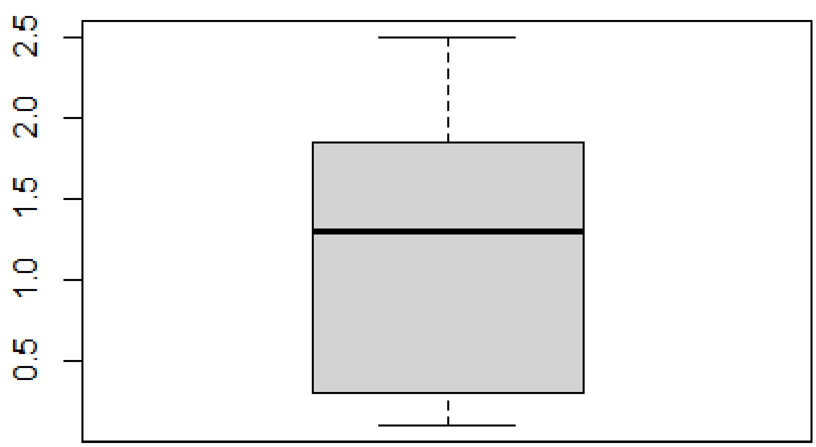

```
iris <- iris %>%
  mutate(Sepal.Width=ifelse(Sepal.Width>10, Sepal.Width/10, Sepal.Width))

filter(iris, Sepal.Width < 1)

## # A tibble: 2 × 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##          <dbl>       <dbl>        <dbl>       <dbl> <chr>
## 1          5            -3          3.5         1   versicolor
## 2          5.7           0          1.7         0.3 setosa

#The negative should be positive
iris <- iris %>%
  mutate(Sepal.Width=ifelse(Sepal.Width<0, Sepal.Width*(-1), Sepal.Width))
#For the zero, everything else seems to be good
#Replace by the mean
iris <- iris %>%
  mutate(Sepal.Width=ifelse(Sepal.Width==0, mean(Sepal.Width, na.rm=TRUE),
Sepal.Width))

#Remaining outliers can be simply big ones.
```

```r
# 3 outliers in Petal.Length
boxplot(iris$Petal.Length)
```




```r
filter(iris, Petal.Length > 10)

## # A tibble: 3 × 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##          <dbl>       <dbl>        <dbl>       <dbl> <chr>
## 1          7.3         2.9           63          NA   virginica
## 2          6.6         2.9           23         1.3 versicolor
## 3          4.9         3             14         2   setosa

iris <- iris %>%
  mutate(Petal.Length=ifelse(Petal.Length>10, Petal.Length/10, Petal.Length))

#It is OK, but the zero!

filter(iris, Petal.Length < 1)

## # A tibble: 3 × 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##          <dbl>       <dbl>        <dbl>       <dbl> <chr>
## 1         NA           2.8        0.82          1.3 versicolor
## 2          5.1         3.8        0             0.2 setosa
## 3          5.5        NA          0.925         1   versicolor

#Even three problems...
# 0 is incorrect, other values are OK, replace by the mean
# 0.82: impossible to know what happened, as Sepal.Length is missing, remove
it
# 0.925: impossible to know what happened, as Sepal.Width is missing, remove
it

iris <- iris %>%
  mutate(Petal.Length=ifelse(Petal.Length==0, mean(Petal.Length, na.rm=TRUE),
Petal.Length))
```

```r
iris <- iris[iris$Petal.Length>1|is.na(iris$Petal.Length),]

summary(iris)

##   Sepal.Length    Sepal.Width     Petal.Length    Petal.Width
##  Min.   :4.300   Min.   :2.200   Min.   :1.100   Min.   :0.10
##  1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.30
##  Median :5.750   Median :3.000   Median :4.450   Median :1.30
##  Mean   :5.819   Mean   :3.062   Mean   :3.857   Mean   : Inf
##  3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.85
##  Max.   :7.900   Max.   :4.200   Max.   :6.900   Max.   : Inf
##  NA's   :9       NA's   :15      NA's   :19      NA's   :12
##    Species
##  Length:147
##  Class :character
##  Mode  :character
##

# No outliers in Petal.Width, but an Inf! Still an outlier.
boxplot(iris$Petal.Width)

## Warning in bplt(at[i], wid = width[i], stats = z$stats[, i], out =
## z$out[z$group == : Outlier (Inf) in boxplot 1 is not drawn
```




```r
filter(iris, Petal.Width > 10)

## # A tibble: 1 × 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##          <dbl>       <dbl>        <dbl>       <dbl> <chr>
## 1          5.1         3.8          1.9         Inf setosa

iris <- iris %>%
  mutate(Petal.Width=ifelse(Petal.Width==Inf, NA, Petal.Width))
```

```r
summary(iris)

##   Sepal.Length    Sepal.Width     Petal.Length    Petal.Width   
##  Min.   :4.300   Min.   :2.200   Min.   :1.100   Min.   :0.100  
##  1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300  
##  Median :5.750   Median :3.000   Median :4.450   Median :1.300  
##  Mean   :5.819   Mean   :3.062   Mean   :3.857   Mean   :1.214  
##  3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800  
##  Max.   :7.900   Max.   :4.200   Max.   :6.900   Max.   :2.500  
##  NA's   :9       NA's   :15      NA's   :19      NA's   :13     
##    Species         
##  Length:147        
##  Class :character  
##  Mode  :character  
## 


# c) Propose a solution for the NAs
# using the information in the dataset only.
# Implement your proposal.

summary(iris)

##   Sepal.Length    Sepal.Width     Petal.Length    Petal.Width   
##  Min.   :4.300   Min.   :2.200   Min.   :1.100   Min.   :0.100  
##  1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300  
##  Median :5.750   Median :3.000   Median :4.450   Median :1.300  
##  Mean   :5.819   Mean   :3.062   Mean   :3.857   Mean   :1.214  
##  3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800  
##  Max.   :7.900   Max.   :4.200   Max.   :6.900   Max.   :2.500  
##  NA's   :9       NA's   :15      NA's   :19      NA's   :13     
##    Species         
##  Length:147        
##  Class :character  
##  Mode  :character  
## 


# Are they in the same rows or not?
filter(iris, is.na(Sepal.Length))

## # A tibble: 9 × 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species   
##          <dbl>       <dbl>        <dbl>       <dbl> <chr>     
## 1           NA         3.9          1.7         0.4 setosa    
## 2           NA         4           NA           0.2 setosa    
## 3           NA         3            5.9         2.1 virginica 
## 4           NA         2.9          4.5         1.5 versicolor
## 5           NA         3.2          5.7         2.3 virginica 
## 6           NA         3.3          5.7         2.1 virginica 
## 7           NA         3            5.5         2.1 virginica 
## 8           NA         2.8          4.7         1.2 versicolor
## 9           NA         3            4.9         1.8 virginica
```

```r
filter(iris, is.na(Sepal.Width))
```

```
## # A tibble: 15 × 5
##    Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##           <dbl>       <dbl>        <dbl>       <dbl> <chr>
##  1          6.2          NA          5.4         2.3 virginica
##  2          5.3          NA         NA           0.2 setosa
##  3          5.5          NA          4           1.3 versicolor
##  4          5.6          NA          4.2         1.3 versicolor
##  5          5.4          NA          4.5         1.5 versicolor
##  6          5            NA          1.2         0.2 setosa
##  7          5.7          NA         NA           0.4 setosa
##  8          6.5          NA          4.6         1.5 versicolor
##  9          4.9          NA          3.3         1   versicolor
## 10          6.7          NA          5           1.7 versicolor
## 11          6.3          NA          4.4         1.3 versicolor
## 12          5.8          NA          5.1         2.4 virginica
## 13          4.8          NA          1.9         0.2 setosa
## 14          5            NA          1.4         0.2 setosa
## 15          5.5          NA          3.8         1.1 versicolor
```

```r
filter(iris, is.na(Petal.Length))
```

```
## # A tibble: 19 × 5
##    Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##           <dbl>       <dbl>        <dbl>       <dbl> <chr>
##  1          5.3        NA             NA         0.2 setosa
##  2         NA           4             NA         0.2 setosa
##  3          4.9         3.6           NA         0.1 setosa
##  4          6.2         2.8           NA         1.8 virginica
##  5          4.4         3.2           NA         0.2 setosa
##  6          6.2         2.9           NA         1.3 versicolor
##  7          5.7        NA             NA         0.4 setosa
##  8          4.9         3.1           NA         0.2 setosa
##  9          5.1         2.5           NA         1.1 versicolor
## 10          5.1         3.5           NA        NA   setosa
## 11          6.9         3.1           NA         1.5 versicolor
## 12          7.2         3.6           NA         2.5 virginica
## 13          5.1         3.7           NA         0.4 setosa
## 14          6.1         2.8           NA         1.3 versicolor
## 15          6.3         3.4           NA         2.4 virginica
## 16          5.9         3             NA         1.5 versicolor
## 17          4.6         3.6           NA         0.2 setosa
## 18          7.7         3             NA         2.3 virginica
## 19          6.4         3.1           NA         1.8 virginica
```

```r
filter(iris, is.na(Petal.Width))

## # A tibble: 13 × 5
##    Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##           <dbl>       <dbl>        <dbl>       <dbl> <chr>
##  1          6.4         2.7          5.3          NA virginica
##  2          6           3            4.8          NA virginica
##  3          7.3         2.9          6.3          NA virginica
##  4          5.9         3.2          4.8          NA versicolor
##  5          6.4         2.8          5.6          NA virginica
##  6          5           2.3          3.3          NA versicolor
##  7          5.1         3.5         NA            NA setosa
##  8          4.9         2.5          4.5          NA virginica
##  9          5.1         3.8          1.9          NA setosa
## 10          6.3         2.7          4.9          NA virginica
## 11          4.8         3            1.4          NA setosa
## 12          4.4         3            1.3          NA setosa
## 13          5.8         2.6          4            NA versicolor

# As no NA in the Species column, we can use this information.
# A possible strategy:
# - delete rows where there ar two missing values
# - replace the only missing value by the mean of the species

iris$missing <- is.na(iris$Sepal.Length)+
      is.na(iris$Sepal.Width)+
      is.na(iris$Petal.Length)+
      is.na(iris$Petal.Width)

iris <- iris %>% filter(missing<2)

#To check if it is OK.
summary(iris$missing)

##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.0000  0.0000  0.0000  0.3357  1.0000  1.0000

# Load the tidyverse library
library(tidyverse)


# Replace missing values with the mean of the column for the given species
cleaned_iris <- iris %>%
  group_by(Species) %>% # Group by Species
  mutate(across(everything(), ~ ifelse(is.na(.), mean(., na.rm = TRUE), .)))
%>%
  select(!missing)

# you can also calculate the mean of each group and species
# and use it in 12 mutates
# mean(iris$Sepal.Length[iris$Species=="setosa"], na.rm = TRUE)
```

```
#Exercise 6.2
#a) Load the “olympics.txt” file. Figure out the file type.

olympics <- read_tsv("olympics.txt")

## New names:
## Rows: 6 Columns: 16
## ── Column specification
## ──────────────────────────────────────────────────────── Delimiter: "\t" chr
## (16): Year, 1980, 1984, 1988, 1992, 1996, 2000, 2004...8, 2004...9, 2004...
## ℹ Use `spec()` to retrieve the full column specification for this data. ℹ
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## • `2004` -> `2004...8`
## • `2004` -> `2004...9`
## • `2004` -> `2004...10`
## • `2004` -> `2004...11`

#b) The data is not tidy. Make it tidy by pivot transformations.

olympics <- olympics %>%
  pivot_longer(!Year, names_to = "year" , values_to="value") %>%
  pivot_wider(names_from = Year, values_from = value)

#c) Check for duplicates. Eliminate them.
olympics <- olympics %>%
  separate(col=year, into=c('year', 'mess'), sep=4) %>%
  select(!mess) %>%
  mutate(across(c(1, 3:7), as.integer))

olympics <- unique(olympics)

glimpse(olympics)

## Rows: 13
## Columns: 7
## $ year                           <int> 1980, 1984, 1988, 1992, 1996, 2000, …
## $ City                           <chr> "Moscow, USSR", "Los Angeles, USA", …
## $ `Length (Days)`                <int> 16, 16, 16, 16, 17, 17, 17, 17, 17, …
## $ `Participating Countries`      <int> 80, 140, 159, 169, 197, 199, 201, 20…
## $ `Participating Athletes`       <int> 5179, 6829, 8391, 9356, 10318, 10651…
## $ `Total Gold Medals`            <int> 7, 15, 10, 14, 16, 20, 22, 22, 25, 3…
## $ `Gold Medals by African Athletes` <int> 204, 221, 237, 257, 271, 300, 301, 3…

olympics <- olympics %>%
  filter(!`Gold Medals by African Athletes`==3001) %>%
  filter(!(year==2020))

#d) Check for outliers, propose a solution.

#Partially solved in filtering for duplicates
olympics <- olympics %>%
  mutate(`Length (Days)`=ifelse(`Length (Days)`==170, 17, `Length (Days)`))
```

```
#e) Check the coherence between variable names and content, solve the
problem.
#Total gold medals and won by African athletes are mixed up.
olympics <- olympics %>%
  rename(Total_Gold_Medals = `Gold Medals by African Athletes`,
         Gold_Medals_by_African_Athletes = `Total Gold Medals`)

#Exercise 6.3
library(tidyverse)
data <- read_csv("immunization.csv")
```

```
## Rows: 12808 Columns: 8
## ── Column specification ────────────────────────────────────────────────────────
## Delimiter: ","
## chr (8): Time, Time Code, Country Name, Country Code, Mortality rate, under-...
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
data2 <- data %>% head(-5)
data2[data2 == ".."] <- NA
data3 <- data2 %>% mutate(year = as.numeric(`Time`),
         c = `Country Code`,
         pop = as.numeric(`Population, total [SP.POP.TOTL]`)/ 1000000,
         mort = as.numeric(`Mortality rate, under-5 (per 1,000 live births)
[SH.DYN.MORT]`),
         surv = (1000 - mort) / 100,
         imm = as.numeric(`Immunization, measles (% of children ages 12-23
months) [SH.IMM.MEAS]`),
         gdppc = as.numeric(`GDP per capita, PPP (constant 2011 international
$) [NY.GDP.PCAP.PP.KD]`),
         lngdppc = log(gdppc)) %>% rename(countryname = "Country Name",
countrycode = "Country Code")

data4 <- data3 %>%
select(c("year","countryname","countrycode","pop","mort","surv","imm","gdppc"
,"lngdppc",)) %>%
  filter(year >= 1998)
data5 <- data4 %>% drop_na(c("imm", "mort", "pop"))
```

```r
summary(data5)

##       year      countryname         countrycode              pop          
##  Min.   :1998   Length:3807        Length:3807        Min.   :   0.0093  
##  1st Qu.:2003   Class :character   Class :character   1st Qu.:   1.7568  
##  Median :2008   Mode  :character   Mode  :character   Median :   7.3956  
##  Mean   :2008                                         Mean   :  35.0675  
##  3rd Qu.:2013                                         3rd Qu.:  23.5769  
##  Max.   :2017                                         Max.   :1386.3950  
##                                                                           
##       mort             surv            imm            gdppc         
##  Min.   :  2.10   Min.   :7.572   Min.   : 8.0   Min.   :   275.5  
##  1st Qu.:  9.30   1st Qu.:9.342   1st Qu.:79.0   1st Qu.:  3114.3  
##  Median : 23.30   Median :9.767   Median :92.0   Median :  9386.1  
##  Mean   : 42.98   Mean   :9.570   Mean   :85.3   Mean   : 16337.0  
##  3rd Qu.: 65.75   3rd Qu.:9.907   3rd Qu.:96.0   3rd Qu.: 22449.1  
##  Max.   :242.80   Max.   :9.979   Max.   :99.0   Max.   :124024.6  
##                                                  NA's   :165       
##     lngdppc      
##  Min.   : 5.619  
##  1st Qu.: 8.044  
##  Median : 9.147  
##  Mean   : 9.042  
##  3rd Qu.:10.019  
##  Max.   :11.728  
##  NA's   :165     

number <- length(unique(data5$countryname))
miss_in <- sum(is.na(data5$gdppc))

number

## [1] 192

miss_in

## [1] 165

data6 <- data5 %>% select(c("year", "countrycode", "imm", "surv")) %>%
arrange(year,countrycode) %>%
  pivot_wider(names_from = c(countrycode),values_from=c(imm,surv))
```